

Důl. 1.

$$(1.1) \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) =$$

$$\frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) = \frac{1}{4} \cdot (\langle u+v, u+v \rangle - \langle u-v, u-v \rangle) =$$

$$\frac{1}{4} \cdot (\langle u+v, u \rangle + \langle u+v, v \rangle - \langle u-v, u \rangle + \langle u-v, v \rangle) =$$

$$\frac{1}{4} \cdot (\overline{\langle u, u+v \rangle} + \overline{\langle v, u+v \rangle} - \overline{\langle u, u-v \rangle} + \overline{\langle v, u-v \rangle}) =$$

$$\frac{1}{4} \cdot (\overline{\langle u, u \rangle} + \overline{\langle u, v \rangle} + \overline{\langle v, u \rangle} + \overline{\langle v, v \rangle} - \overline{\langle u, u \rangle} + \overline{\langle u, v \rangle} + \overline{\langle v, u \rangle} - \overline{\langle v, v \rangle}) =$$

$$\frac{1}{4} \cdot (2 \cdot \overline{\langle u, v \rangle} + 2 \cdot \overline{\langle v, u \rangle}) = \frac{1}{2} \cdot (\overline{\langle u, v \rangle} + \langle u, v \rangle) = \frac{1}{2} \cdot 2 \operatorname{Re}(\langle u, v \rangle) =$$

Proble:  $x + \bar{x} = 2 \operatorname{Re}(x)$  ↑

$\operatorname{Re}(\langle u, v \rangle)$  ← a to jsme chtěli dokázat!

Rozepsal jsem si druhou část rovnice.

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$= \frac{1}{4} \cdot (\langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle) = \frac{1}{4} \cdot (\langle u-iv, u \rangle + \langle u-iv, -iv \rangle - (\langle u+iv, u \rangle + \langle u+iv, iv \rangle)) =$$

$$\langle u+iv, iv \rangle) = \frac{1}{4} \cdot (\overline{\langle u, u-iv \rangle} + \overline{\langle iv, u-iv \rangle} - (\overline{\langle u, u+iv \rangle} + \overline{\langle iv, u+iv \rangle})) =$$

$$\frac{1}{4} \cdot (\overline{\langle u, u \rangle} + \overline{\langle u, -iv \rangle} + \overline{\langle iv, u \rangle} + \overline{\langle iv, -iv \rangle} - (\overline{\langle u, u \rangle} + \overline{\langle u, +iv \rangle} + \overline{\langle iv, u \rangle} + \overline{\langle iv, iv \rangle})) =$$

$$\frac{1}{4} \cdot (\overline{\langle u, u \rangle} + \overline{\langle u, -iv \rangle} + \overline{\langle iv, u \rangle} + \overline{\langle iv, -iv \rangle} - \overline{\langle u, u \rangle} - \overline{\langle u, +iv \rangle} - \overline{\langle iv, u \rangle} - \overline{\langle iv, iv \rangle}) =$$

$$\frac{1}{4} \cdot ((-i) \cdot \overline{\langle u, v \rangle} + (-i) \cdot \overline{\langle u, v \rangle} + (i) \cdot (i) \cdot \overline{\langle v, v \rangle} - (i) \cdot \overline{\langle u, v \rangle} - i \cdot \overline{\langle u, v \rangle} - (i) \cdot (i) \cdot \overline{\langle v, v \rangle}) =$$

$$\frac{1}{4} \cdot (i \cdot \overline{\langle u, v \rangle} - i \cdot \overline{\langle u, v \rangle} + \overline{\langle v, v \rangle} + i \cdot \overline{\langle u, v \rangle} - i \cdot \overline{\langle u, v \rangle} - \overline{\langle v, v \rangle}) =$$

$$\frac{1}{4} \cdot (i \cdot \overline{\langle u, v \rangle} - i \langle u, v \rangle + \overline{\langle v, v \rangle} + i \cdot \overline{\langle u, v \rangle} - i \langle u, v \rangle - \overline{\langle v, v \rangle}) =$$

$$\frac{1}{4} \cdot (2 \cdot i \cdot \overline{\langle u, v \rangle} - 2i \cdot \langle u, v \rangle) = \frac{2i}{4} \cdot (\overline{\langle u, v \rangle} - \langle u, v \rangle) \stackrel{*}{=} \frac{2i}{4} \cdot (-2i \cdot \operatorname{Im}(\langle u, v \rangle)) =$$

$$\stackrel{*}{*} (\overline{x} - x) = (\overline{a+bi}) - (a+bi) = (a-bi) - (a+bi) = a-bi-a-bi = -2i \cdot b = -2i \cdot \operatorname{Im}(x)$$

$$\frac{4 \cdot i^2}{4} \cdot \operatorname{Im}(\langle u, v \rangle) = \operatorname{Im}(\langle u, v \rangle)$$

Levá strana je rovna pravé, tím jsem dokázal rovnost.

Rovnost musí určitě platit, protože operace, které jsem používal jsem definován pro skalární součin.